



Medreach

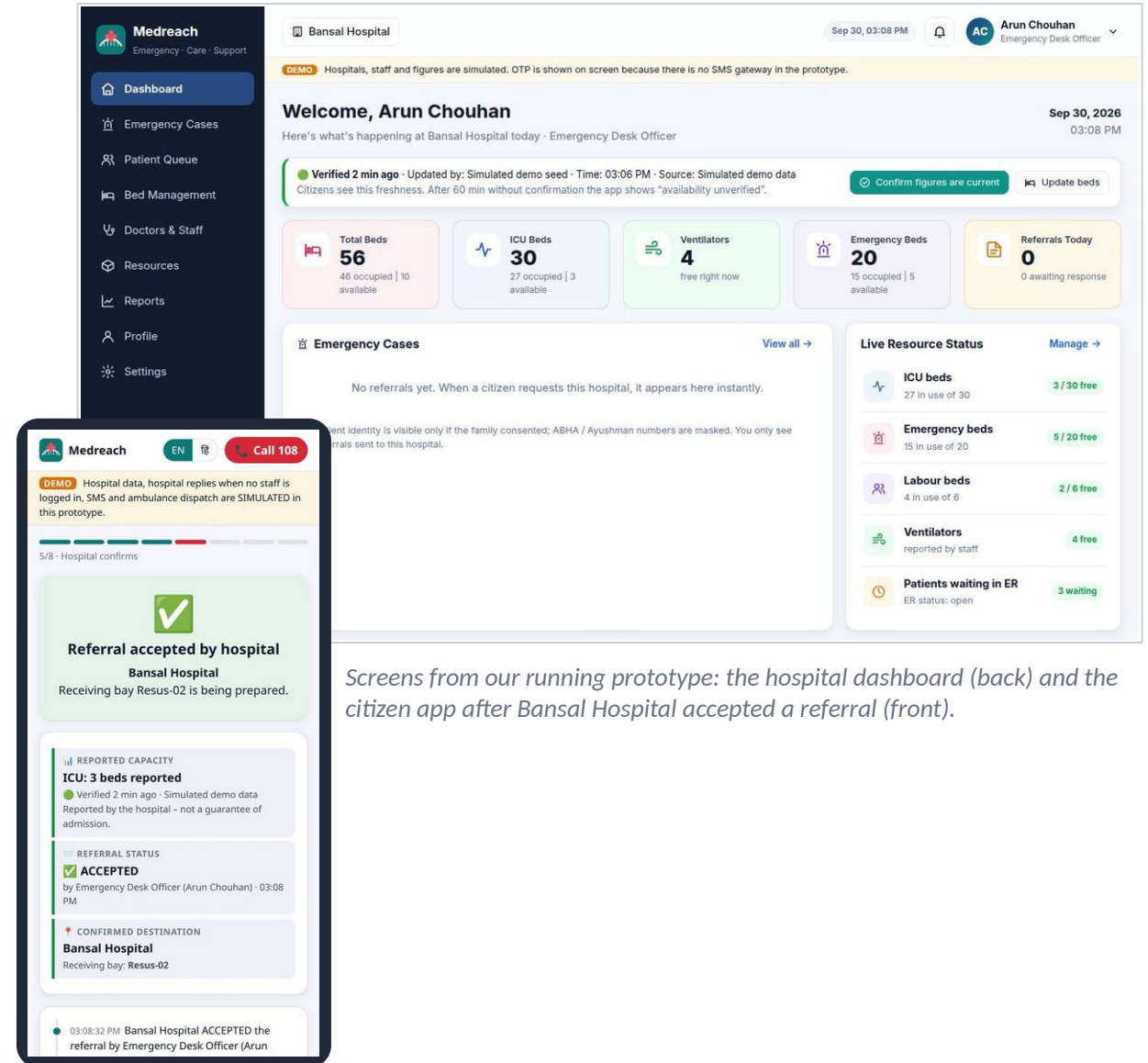
MPOne Idea & Innovation Hackathon 2026 • Challenge 5: AI for Public Services, Healthcare

Get the right hospital to say yes before the patient leaves home

Medreach is a working emergency-coordination system for Madhya Pradesh. A family describes the emergency in Hindi or English. We find a hospital that can treat it right now, the hospital accepts and prepares a bed, and the ambulance is tracked until arrival.

Live prototype medreach-wtne.onrender.com

Source code github.com/prakharpurwal518-collab/Medreach



Screens from our running prototype: the hospital dashboard (back) and the citizen app after Bansal Hospital accepted a referral (front).

What Medreach does, and who it is for

Medreach is an emergency coordination platform, not a diagnosis app. It joins three people who today work blind: the family in panic, the hospital emergency desk, and the health department.

Citizens and families

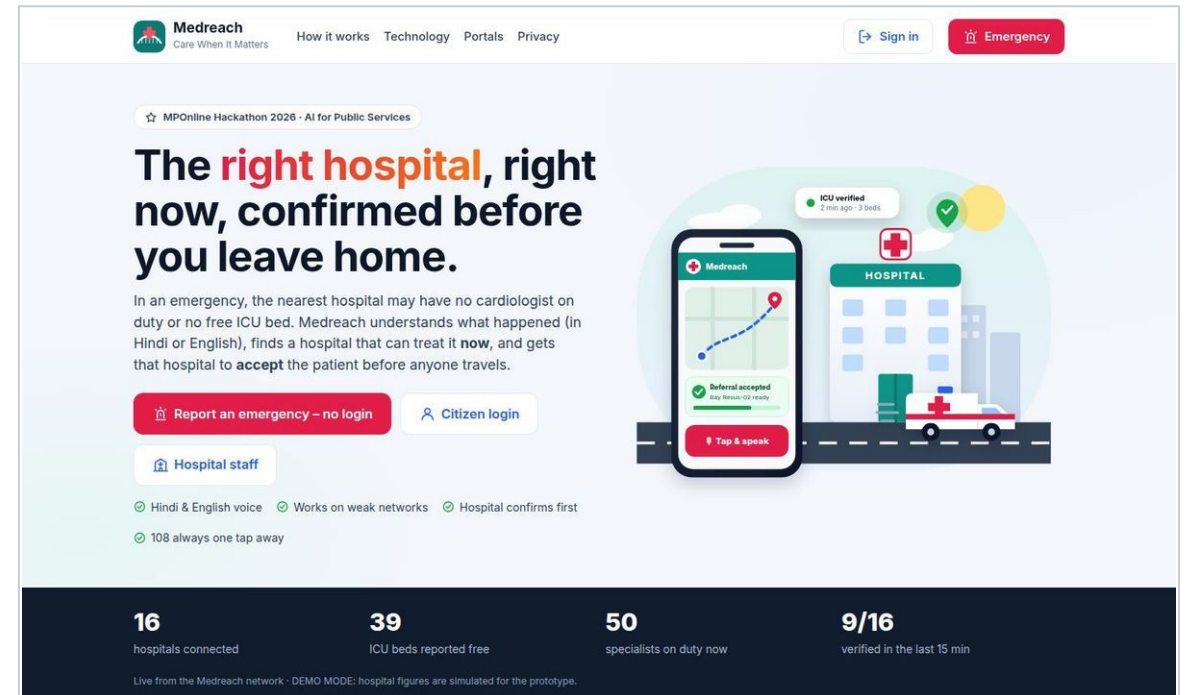
Speak the emergency, see only hospitals that can treat it, get a confirmed bed and a tracking link for relatives. No login needed. “I’m alone” mode for people living alone.

Hospital emergency desks

A verified board of beds, ICU, ventilators and doctors on duty. Referrals arrive with an AI-written pre-arrival (SBAR) note; one tap to accept and assign a bay.

Health department, Govt. of MP

A live, audited view of emergency capacity across hospitals, and data on where patients are refused or delayed.



Our home page, with live network figures (demo data is labelled as simulated).

Live

deployed and
running on the web

2

private portals

16

Bhopal-region
hospitals modelled

7 of 7

suggested
technologies used

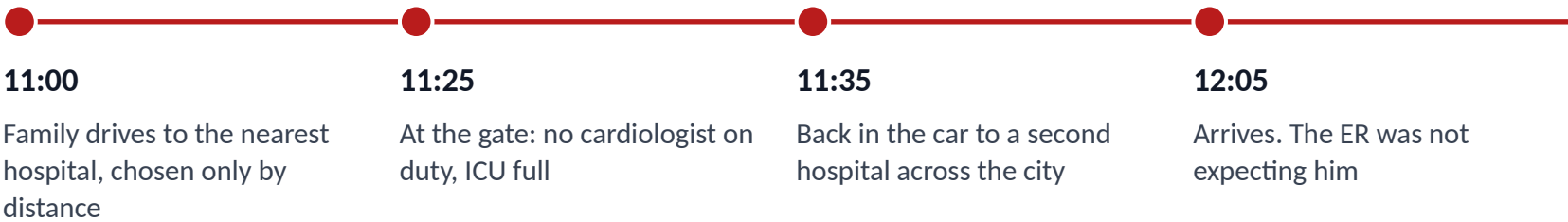
51

automated tests
passing

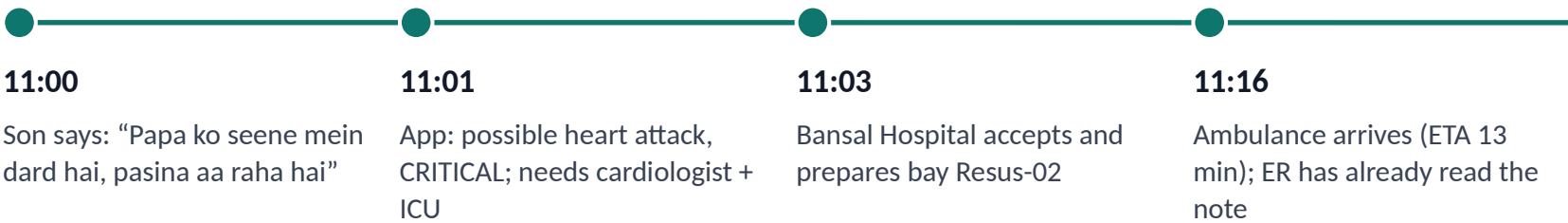
11 pm in Bhopal: one emergency, two outcomes

A 62-year-old man has chest pain and is sweating. His son has a phone and a car.

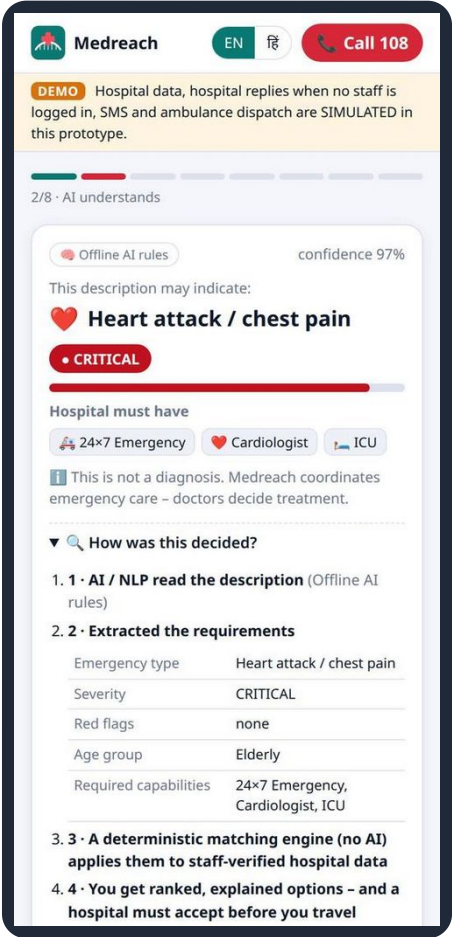
Today



With Medreach



One trip instead of two, and a team that is ready on arrival. Times are illustrative; the 13-minute ETA and bay come from our prototype run.



What the son sees about a minute after speaking.

Why patients lose the golden hour

1,68,491

people died in road accidents in India in 2022

Source: Ministry of Road Transport & Highways, “Road Accidents in India 2022”.

Road crashes are only one part. Heart attacks, strokes, burns and childbirth emergencies face the same problem: the first hospital reached is often not equipped for that case at that hour, and nobody finds out until the family arrives.

01 Nearest is not capable

Maps show distance, not whether the cardiologist is on duty, the ICU has a bed or the CT scanner works tonight.

02 No trusted live status

Bed figures, where they exist, are old and have no owner or time stamp, so nobody trusts them.

03 No acceptance before travel

Families arrive unannounced. Refusal happens at the gate, after the journey.

04 Panic and language

In an emergency people cannot fill forms. Many would rather speak, and in Hindi.

05 People living alone

An elderly person alone cannot call hospitals, arrange transport and inform family at once.

06 Information is lost

Allergies, history and Ayushman / ABHA details do not reach the ER doctor before the patient.

Who feels it

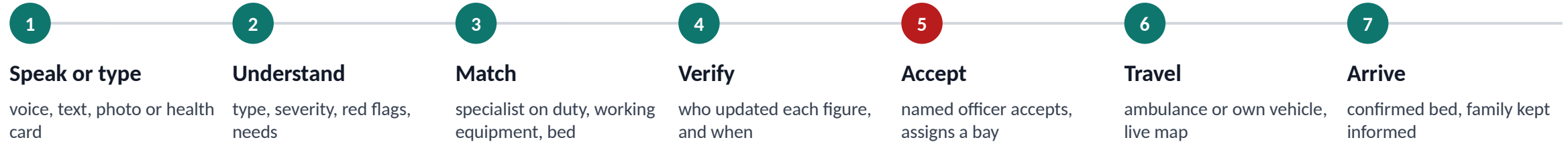
Patients & families
wasted trips, refusals, delayed treatment

Rural citizens & ASHA workers
long distances make a wrong first choice costly

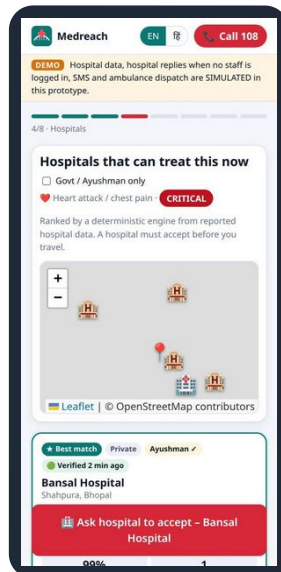
Hospital ER teams
unannounced critical arrivals, crowding

Health department
no live picture of emergency capacity

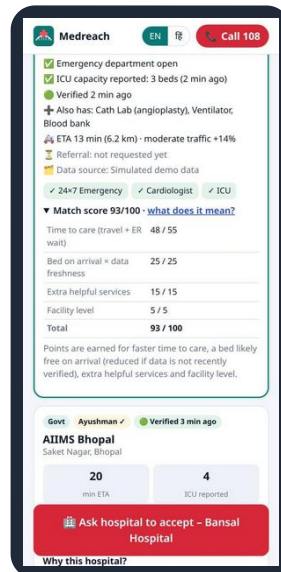
One flow, from the first words to a confirmed bed



Step 1 • Tap and speak, or pick a common emergency



Step 3 • Only hospitals that can treat this case



Step 4 • “Why this hospital?” with each score part

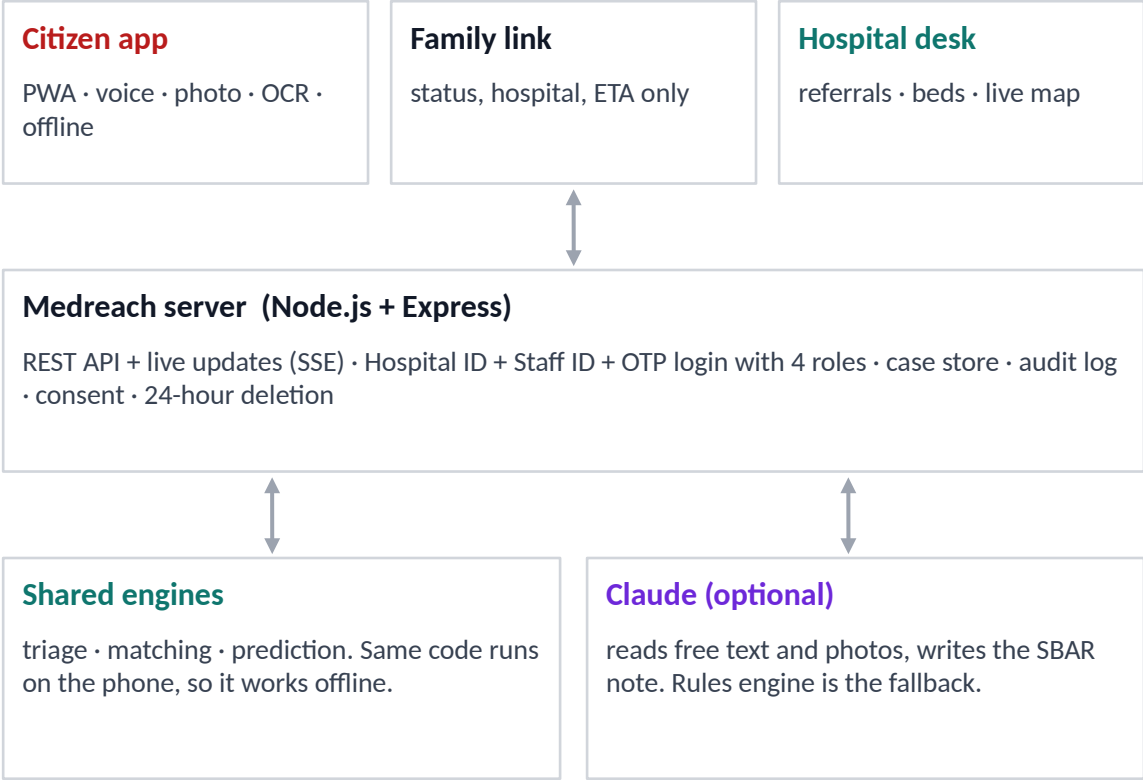


Step 6 • Ambulance status and navigation

Rules we never break

- The hospital accepts before anyone travels.
- Only hospitals that can treat the case are offered.
- No answer in 60 seconds: the next hospital is asked automatically.
- It never diagnoses; it says “may indicate”.
- Call 108 stays on every screen.

How it is built



Live-mode integrations: Health Facility Registry · hospital HMIS · 108 control room · ABDM · SMS gateway

Technology	What it does in Medreach
Generative AI	Turns “papa ko seene mein dard...” into requirements; writes the ER note
Voice bots	Hindi / English speech in; acceptance and directions read aloud
Computer vision	Photo of an injury → visible findings that can raise urgency
OCR	Reads Ayushman / ABHA cards on the phone; shared only with consent
Predictive analytics	Traffic ETA, chance a bed is still free on arrival, 8-hour ER forecast
Conversational AI	Asks one follow-up question at a time (conscious? breathing?)
Mobile app	Installable PWA, bilingual, works on weak networks

Built with: Node.js 22 · Express 5 · PWA + service worker · Leaflet with OpenStreetMap (no API keys) · Tesseract.js · Web Speech API · Claude SDK · 51 automated tests

Ranking: time to care (55 pts) + bed free on arrival × data freshness (25) + helpful services (15) + facility level (5). **Safety:** severity = max(AI, rules); AI never picks the hospital.

From prototype to a state-wide service

Prototype done

Both portals live, AI triage, matching, acceptance, live maps, privacy controls, 51 tests.

Pilot in Bhopal about 3 months

Onboard a group of government and private hospitals, name a nodal officer each, train ER desks, clinical review of triage rules, SMS OTP.

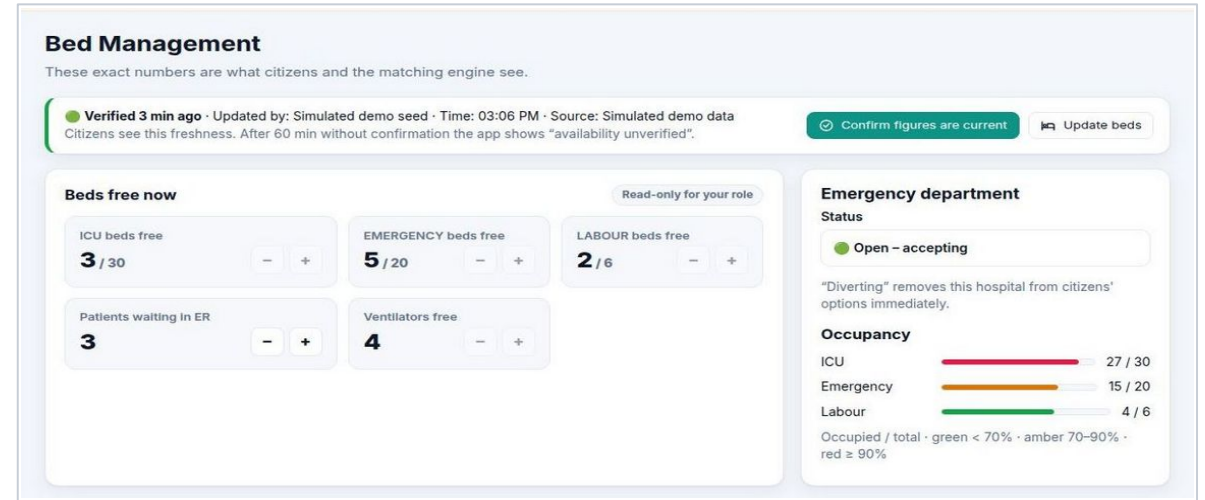
Integrate about 3–6 months

Health Facility Registry, hospital HMIS feeds, 108 control room hand-off, ABDM records, encrypted database.

Scale across MP about 6–12 months

District-by-district rollout, state command view, IVR / missed-call option for feature phones.

Durations are planning estimates and depend on hospital onboarding and approvals.



What a hospital keeps current: beds, ventilators, ER status. One click re-confirms the figures.

Risk	How we handle it
Hospitals stop updating	Every figure shows its age; stale data is ranked lower; one-click confirm
AI misreads a case	AI can only raise urgency; a deterministic engine decides
No hospital answers	After 60 s the next suitable hospital is asked
Weak or no network	Triage and matching run on the phone from saved data

What existing tools do not do

	Map search	Bed portals	Helplines	Medreach
Understands a spoken emergency in Hindi	No	No	Operator	Yes
Matches specialist and equipment, not just beds	No	Beds only	Manual	Yes
Shows who updated each figure, and when	No	Rarely	No	Every figure
Hospital accepts before the family travels	No	No	Sometimes	Yes, with bay
Moves to the next hospital automatically	No	No	Manual	After 60 s
Sends a pre-arrival note to the ER doctor	No	No	No	SBAR note
Handles a person who is alone	No	No	No	"I'm alone"

We also keep three things separate on every screen: what a hospital reported, whether it accepted, and the confirmed destination. A family never mistakes “2 ICU beds” for “admitted”.



- 1 AI-written SBAR note, built from what the son said
- 2 Desk officer assigns a bay and accepts in one tap

< 1 s for a hospital’s status change to reach citizens

60 s before the next hospital is asked

0 API keys needed for maps

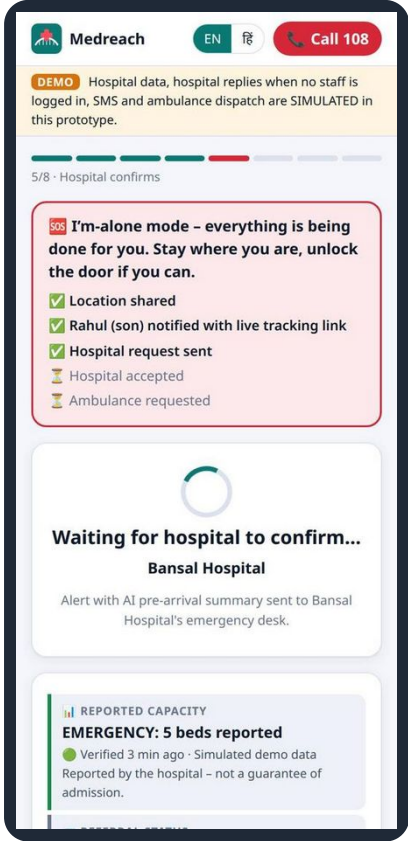
Try it: set a hospital’s ER to “Diverting” and it disappears from every family’s list immediately.

Who gains, and how we will measure it

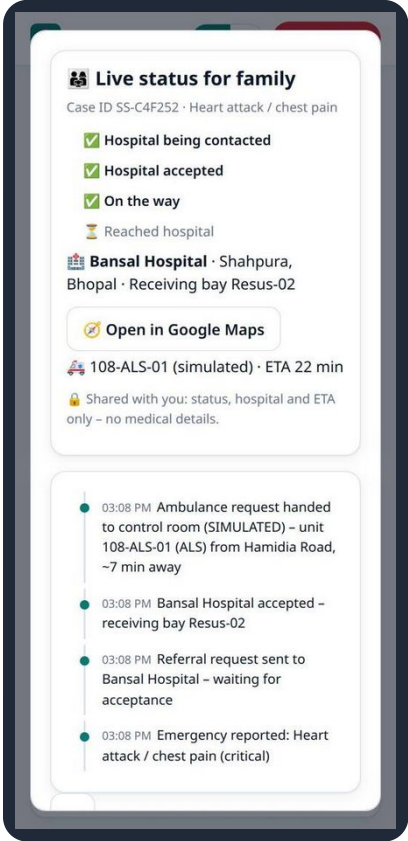
Area	Benefit
Social	Right care the first time; equal access for rural, elderly and Hindi-speaking citizens
Economic	No repeat trips or wasted visits; better use of ICU beds and specialists
Governance	Live, audited view of emergency capacity for planning beds and ambulances
Educational	Safe first-aid steps, clear “not a diagnosis” messaging, awareness of Ayushman, ABHA, 108
User	One button, voice, two languages, clear status and a family tracking link
Health system	Fewer surprise arrivals, less ER crowding, fewer transfers between hospitals

What the pilot will measure

Measure	How we get it
Time to definitive care	From the first request to the treating hospital, from case timestamps
Transfers avoided	Patients who needed a second hospital, before vs during the pilot
Acceptance time	From referral sent to “accepted” in the audit log



“I’m alone”: one tap shares location, alerts a contact, books the hospital and an ambulance.



The family link: status, hospital and ETA. No medical details.

Why Medreach should be selected

Thank you

Reason	Evidence you can check
It solves a real, life-or-death gap	Families in MP lose the golden hour at the wrong hospital; Medreach confirms a bed before travel
It already works	Both portals are live at medreach-wtne.onrender.com; the whole flow takes about 3 minutes to try
All 7 technologies are real	Each one is in working code, listed file by file in our repository
The AI is safe and explainable	AI can only raise urgency; a tested engine decides and shows why; 108 is always on screen
Privacy is built in	Consent before sharing, masked IDs, staff OTP and roles, audit log, deletion 24 h after hand-over
It can scale cheaply	Open-source stack, free maps, one web app for all phones; clear path to HFR, HMIS, 108 and ABDM

Our ask: a 3-month pilot with a group of Bhopal hospitals, followed by integration with the Health Facility Registry, hospital HMIS, the 108 control room and ABDM for a state-wide rollout.

Live prototype: medreach-wtne.onrender.com Code: github.com/prakharpurwal518-collab/Medreach Hospital demo login: Bansal Hospital • BANSAL-ED01